

An explicit graph basis and exact Jacobian certificates for a self-dual five-form in ten dimensions

uskutsav-cpu

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Abstract

We give a reproducible certificate for 81 explicit contractions of a real self-dual five-form in signature $(1, 9)$. The fixed selection consists of one quartic, two sextic, six octic, twelve degree-ten, and sixty degree-twelve polynomials. Their Jacobian has a nonzero 81×81 minor at four exact finite-field specializations, using the same ordered functions in every case. An integral coordinate convention makes each nonzero minor a characteristic-zero independence witness. The construction completes the functional-independence problem, with the known generic quotient dimension as upper bound. It does not determine all polynomial generators or syzygies.

1 Scope and conventions

Let F be an alternating five-form, with metric $\eta = \text{diag}(-1, 1, \dots, 1)$ and Hodge convention $\star F = F$. The self-dual subspace has dimension 126. The literature supplies the generic quotient dimension 81 and the low-degree Hilbert coefficients [1, 2]. The present certificate supplies the matching lower bound using explicit graph contractions. The contraction catalogs and previously discovered bases are inherited from the repository's research branches; the certificate reported here re-evaluates those functions at new points.

A graph has one vertex per copy of F , no loops, and valence five. An edge of multiplicity m contracts m index pairs with inverse metrics. Vertices and edges are ordered; slot order is fixed by lexicographic edge order. Graph isomorphism alone does not fix the sign of an odd-form contraction, so the formula identifier hashes the ordered edge record. Disconnected contractions are products. The quadratic contraction vanishes on this self-dual space.

For each increasing five-tuple I containing 0, define $A_I = F_I$ and let e_I denote its compact alternating component vector. The integral parametrization is

$$F(A) = \sum_{I \ni 0} A_I (e_I + \star e_I).$$

The 126 columns contain only 0, 1, -1 ; their components indexed by $I \ni 0$ form an identity matrix. Every graph polynomial therefore has integer coefficients in A , with no projector denominators.

2 Fixed selection and exact witness

The nine low-degree functions are those in `results/10d_order8.json`; the next twelve come from `results/10d_order10.json`. The first sixty selected functional directions in `results/10d_`

`order12.json` complete the list. The remaining two degree-twelve primitive polynomial directions belong to the larger homogeneous inventory and are not included in the 81-function Jacobian. A dependent Jacobian row does not identify a polynomial syzygy.

Field degree	New functional directions	Cumulative rank
4	1	1
6	2	3
8	6	9
10	12	21
12	60	81

Write $J_{aI} = \partial G_a / \partial A_I$. Each certificate cell records the prime, all 126 coordinates, all 252 self-dual compact components, the same ordered 81 graph identifiers, all entries of the 81×126 Jacobian, 81 pivot columns, and the determinant of that minor.

Prime	Seed	Rank	$\det J_* \bmod p$
32749	20260907	81	20345
32749	20260908	81	30761
32719	20260907	81	1653
32719	20260908	81	2167

The determinant is recomputed by Python-integer elimination independently of the streaming rank sieve. Euler’s identity $\sum_I A_I J_{aI} = \deg(G_a) G_a$ is checked for every row.

If a saved determinant is nonzero modulo p , the corresponding integer determinant polynomial is not identically zero. Its nonvanishing defines a nonempty Zariski-open set in characteristic zero on which these 81 polynomials are algebraically independent. Invariance and the known generic quotient dimension supply the matching upper bound. These are generic local coordinates; neither global orbit separation nor polynomial generation of the entire ring follows.

3 Polynomial spaces and literature comparison

The homogeneous-space dimensions through degree twelve are 1, 2, 7, 14, 72. Products must be included in value-space comparisons. At degree twelve the ten product monomials are I_4^3 , the three symmetric products of sextics, and the six products $I_4 I_{8,j}$. The inherited degree-twelve computation reaches rank 72 using these products and 62 graph directions. Polynomial-space evidence uses gradients concatenated at several points; the functional Jacobian uses one point. These are separate sieves. Factor exponents in a Hilbert product are not automatically generator counts at higher degrees when generators and relations may coexist.

The octic comparison uses a selection of six displayed literature contractions: the first five expressions from equations (4.12)–(4.15) of [1], together with the hatted expression (4.18). The exact machine-readable map includes the quartic-square product explicitly. The 6×6 block compares primitive classes modulo products; any product correction is retained in the full 7×7 matrix.

At degree ten the repository’s implemented readings of the twelve published candidates span a twelve-dimensional space whose intersection with the two-dimensional product space has dimension one in the reconstructed coordinate matrix. Thus a claimed invertible 12×12 identification with the graph primitive complement would be incorrect for those readings. The delivered map has twelve rows and fourteen atlas columns and records the source-reading qualifications. Rational

reconstruction and fresh modular holdouts are strong checks on these maps, but finite samples alone do not prove a characteristic-zero polynomial identity. These qualifications do not affect the graph-only nonzero-minor proof.

4 Reproduction and validation

Use the declared dependencies and run the regression suite before evaluating certificates. The original low-degree commit is preserved by `v0.1-low-degree-complete`. The release commands are:

```
python scripts/graph_to_latex.py
python scripts/search_rank81.py
python scripts/verify_rank81.py
python scripts/verify_rank81.py --recompute
python scripts/validate_rank81_lorentz.py
python scripts/map_literature_basis.py --degree 8
python scripts/map_literature_basis.py --degree 10
```

The default verifier checks the saved matrix and metadata; the explicit `--recompute` mode repeats the graph evaluations. These modes are reported separately. The search stage checkpoints each raw row and its pivot, checks run identity on resume, and bounds worker submissions. Degree-specific discovery remains available through the existing nauty pipelines. `run_degree.py` accepts additional edge-list JSONL streams and separates polynomial completeness from functional rank.

Regression tests cover the 6D reproduction, self-duality, the zero quadratic contraction, exact integer arithmetic, reverse Jacobians against an independent oracle, graph parsing beyond ten vertices, and genuine Lorentz boosts. The final graph selection is checked under both a spatial rotation and a boost; neither uses floating-point tolerances.

References

- [1] M. Cederwall, J. Hutomo, S. M. Kuzenko, K. Lechner, and D. P. Sorokin, *Some remarks on invariants*, [arXiv:2509.14350v2](#).
- [2] J. Hutomo, K. Lechner, and D. P. Sorokin, *On non-linear chiral 4-form theories in D=10*, [arXiv:2509.14351](#).

A Ordered Einstein-index formulas

Each repeated index labels exactly one graph edge; its upper and lower occurrences supply one inverse metric. The machine-readable companion `results/rank81_basis.json` gives the edge lists, adjacency matrices, formula hashes, and contraction costs.

I4_1

$$F_{\mu_0,2,1\mu_0,2,2\mu_0,2,3\mu_0,3,1\mu_0,3,2} F_{\mu_1,2,1\mu_1,2,2\mu_1,3,1\mu_1,3,2\mu_1,3,3} \\ F^{\mu_0,2,1\mu_0,2,2\mu_0,2,3\mu_1,2,1\mu_1,2,2} F^{\mu_0,3,1\mu_0,3,2\mu_1,3,1\mu_1,3,2\mu_1,3,3}$$

I6_1

$$F_{\mu_0,3,1\mu_0,3,2\mu_0,3,3\mu_0,4,1\mu_0,4,2} F_{\mu_1,3,1\mu_1,3,2\mu_1,5,1\mu_1,5,2\mu_1,5,3} \\ F_{\mu_2,4,1\mu_2,4,2\mu_2,4,3\mu_2,5,1\mu_2,5,2} F^{\mu_0,3,1\mu_0,3,2\mu_0,3,3\mu_1,3,1\mu_1,3,2} \\ F^{\mu_0,4,1\mu_0,4,2\mu_2,4,1\mu_2,4,2\mu_2,4,3} F^{\mu_1,5,1\mu_1,5,2\mu_1,5,3\mu_2,5,1\mu_2,5,2}$$

I6_2

$$F_{\mu_0,2,1\mu_0,2,2\mu_0,3,1\mu_0,3,2\mu_0,5,1} F_{\mu_1,3,1\mu_1,3,2\mu_1,4,1\mu_1,4,2\mu_1,5,1} \\ F_{\mu_2,4,1\mu_2,4,2\mu_2,5,1} F^{\mu_0,3,1\mu_0,3,2\mu_1,3,1\mu_1,3,2} \\ F^{\mu_1,4,1\mu_1,4,2\mu_2,4,1\mu_2,4,2} F^{\mu_0,5,1\mu_1,5,1\mu_2,5,1\mu_3,5,1\mu_4,5,1}$$

I8_1

$$F_{\mu_0,4,1\mu_0,4,2\mu_0,4,3\mu_0,4,4\mu_0,5,1} F_{\mu_1,4,1\mu_1,6,1\mu_1,6,2\mu_1,6,3\mu_1,6,4} \\ F_{\mu_2,5,1\mu_2,5,2\mu_2,5,3\mu_2,5,4\mu_2,7,1} F_{\mu_3,6,1\mu_3,7,1\mu_3,7,2\mu_3,7,3\mu_3,7,4} \\ F^{\mu_0,4,1\mu_0,4,2\mu_0,4,3\mu_0,4,4\mu_1,4,1} F^{\mu_0,5,1\mu_2,5,1\mu_2,5,2\mu_2,5,3\mu_2,5,4} \\ F^{\mu_1,6,1\mu_1,6,2\mu_1,6,3\mu_1,6,4\mu_3,6,1} F^{\mu_2,7,1\mu_3,7,1\mu_3,7,2\mu_3,7,3\mu_3,7,4}$$

I8_2

$$F_{\mu_0,3,1\mu_0,3,2\mu_0,3,3\mu_0,3,4\mu_0,6,1} F_{\mu_1,4,1\mu_1,4,2\mu_1,4,3\mu_1,4,4\mu_1,6,1} \\ F_{\mu_2,5,1\mu_2,5,2\mu_2,5,3\mu_2,5,4\mu_2,6,1} F_{\mu_3,7,1}^{\mu_0,3,1\mu_0,3,2\mu_0,3,3\mu_0,3,4} \\ F_{\mu_4,7,1}^{\mu_1,4,1\mu_1,4,2\mu_1,4,3\mu_1,4,4} F_{\mu_5,7,1}^{\mu_2,5,1\mu_2,5,2\mu_2,5,3\mu_2,5,4} \\ F_{\mu_6,7,1}^{\mu_0,6,1\mu_1,6,1\mu_2,6,1} F^{\mu_3,7,1\mu_4,7,1\mu_5,7,1\mu_6,7,1\mu_6,7,2}$$

I8_3

$$F_{\mu_0,3,1\mu_0,3,2\mu_0,3,3\mu_0,3,4\mu_0,4,1} F_{\mu_1,4,1\mu_1,5,1\mu_1,6,1\mu_1,6,2\mu_1,7,1} \\ F_{\mu_2,5,1\mu_2,5,2\mu_2,5,3\mu_2,5,4\mu_2,6,1} F_{\mu_3,7,1}^{\mu_0,3,1\mu_0,3,2\mu_0,3,3\mu_0,3,4} \\ F_{\mu_4,6,1\mu_4,7,1\mu_4,7,2}^{\mu_1,4,1\mu_1,4,1} F^{\mu_1,5,1\mu_2,5,1\mu_2,5,2\mu_2,5,3\mu_2,5,4} \\ F_{\mu_6,7,1}^{\mu_1,6,1\mu_1,6,2\mu_2,6,1\mu_4,6,1} F^{\mu_1,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2\mu_6,7,1}$$

I8_4

$$F_{\mu_0,3,1\mu_0,3,2\mu_0,6,1\mu_0,6,2\mu_0,7,1} F_{\mu_1,4,1\mu_1,4,2\mu_1,4,3\mu_1,4,4\mu_1,6,1} \\ F_{\mu_2,5,1\mu_2,5,2\mu_2,5,3\mu_2,5,4\mu_2,6,1} F_{\mu_3,6,1\mu_3,7,1\mu_3,7,2}^{\mu_0,3,1\mu_0,3,2} \\ F_{\mu_4,7,1}^{\mu_1,4,1\mu_1,4,2\mu_1,4,3\mu_1,4,4} F_{\mu_5,7,1}^{\mu_2,5,1\mu_2,5,2\mu_2,5,3\mu_2,5,4} \\ F^{\mu_0,6,1\mu_0,6,2\mu_1,6,1\mu_2,6,1\mu_3,6,1} F^{\mu_0,7,1\mu_3,7,1\mu_3,7,2\mu_4,7,1\mu_5,7,1}$$

I8_5

$$\begin{aligned}
& F_{\mu_0,3,1\mu_0,3,2\mu_0,4,1\mu_0,6,1\mu_0,6,2} F_{\mu_1,4,1\mu_1,4,2\mu_1,5,1\mu_1,7,1\mu_1,7,2} \\
& F_{\mu_2,5,1\mu_2,5,2\mu_2,5,3\mu_2,5,4\mu_2,6,1} F_{\mu_3,6,1\mu_3,6,2\mu_3,7,1}^{\mu_0,3,1\mu_0,3,2} \\
& F_{\mu_4,7,1\mu_4,7,2}^{\mu_0,4,1\mu_1,4,1\mu_1,4,2} F_{\mu_1,5,1\mu_2,5,1\mu_2,5,2\mu_2,5,3\mu_2,5,4}^{\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_3,6,1\mu_3,6,2} F_{\mu_1,7,1\mu_1,7,2\mu_3,7,1\mu_4,7,1\mu_4,7,2}^{\mu_1,7,1}
\end{aligned}$$

I8_6

$$\begin{aligned}
& F_{\mu_0,3,1\mu_0,3,2\mu_0,6,1\mu_0,6,2\mu_0,7,1} F_{\mu_1,4,1\mu_1,4,2\mu_1,5,1\mu_1,6,1\mu_1,6,2} \\
& F_{\mu_2,4,1\mu_2,5,1\mu_2,5,2\mu_2,7,1\mu_2,7,2} F_{\mu_3,6,1\mu_3,7,1\mu_3,7,2}^{\mu_0,3,1\mu_0,3,2} \\
& F_{\mu_4,5,1\mu_4,5,2}^{\mu_1,4,1\mu_1,4,2\mu_2,4,1} F_{\mu_1,5,1\mu_2,5,1\mu_2,5,2\mu_4,5,1\mu_4,5,2}^{\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_1,6,1\mu_1,6,2\mu_3,6,1} F_{\mu_0,7,1\mu_2,7,1\mu_2,7,2\mu_3,7,1\mu_3,7,2}^{\mu_0,7,1}
\end{aligned}$$

I10_1

$$\begin{aligned}
& F_{\mu_0,4,1\mu_0,4,2\mu_0,4,3\mu_0,7,1\mu_0,9,1} F_{\mu_1,5,1\mu_1,6,1\mu_1,6,2\mu_1,8,1\mu_1,8,2} \\
& F_{\mu_2,5,1\mu_2,5,2\mu_2,7,1\mu_2,7,2\mu_2,9,1} F_{\mu_3,6,1\mu_3,6,2\mu_3,6,3\mu_3,8,1\mu_3,9,1}^{\mu_3,6,1} \\
& F_{\mu_4,8,1\mu_4,8,2}^{\mu_0,4,1\mu_0,4,2\mu_0,4,3} F_{\mu_1,5,1\mu_2,5,1\mu_2,5,2}^{\mu_1,5,1} \\
& F_{\mu_1,6,1\mu_1,6,2\mu_3,6,1\mu_3,6,2\mu_3,6,3} F_{\mu_0,7,1\mu_2,7,1\mu_2,7,2\mu_5,7,1}^{\mu_0,7,1} \\
& F_{\mu_1,8,1\mu_1,8,2\mu_3,8,1\mu_4,8,1\mu_4,8,2} F_{\mu_0,9,1\mu_2,9,1\mu_3,9,1\mu_5,9,1\mu_7,9,1}^{\mu_0,9,1}
\end{aligned}$$

I10_2

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,5,2\mu_0,6,1\mu_0,6,2\mu_0,9,1} F_{\mu_1,5,1\mu_1,5,2\mu_1,5,3\mu_1,8,1\mu_1,9,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,9,1\mu_2,9,2} F_{\mu_3,7,1\mu_3,7,2\mu_3,8,1\mu_3,8,2\mu_3,9,1}^{\mu_3,7,1} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,7,3\mu_4,8,1\mu_4,8,2} F_{\mu_0,5,1\mu_0,5,2\mu_1,5,1\mu_1,5,2\mu_1,5,3}^{\mu_0,5,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_2,6,2\mu_2,6,3} F_{\mu_3,7,1\mu_3,7,2\mu_4,7,1\mu_4,7,2\mu_4,7,3}^{\mu_3,7,1} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_4,8,1\mu_4,8,2} F_{\mu_0,9,1\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_3,9,1}^{\mu_0,9,1}
\end{aligned}$$

I10_3

$$\begin{aligned}
& F_{\mu_0,4,1\mu_0,4,2\mu_0,4,3\mu_0,7,1\mu_0,7,2} F_{\mu_1,5,1\mu_1,6,1\mu_1,6,2\mu_1,6,3\mu_1,9,1} \\
& F_{\mu_2,5,1\mu_2,5,2\mu_2,8,1\mu_2,8,2\mu_2,9,1} F_{\mu_3,6,1\mu_3,7,1\mu_3,7,2\mu_3,8,1\mu_3,9,1}^{\mu_3,6,1} \\
& F_{\mu_4,8,1\mu_4,9,1}^{\mu_0,4,1\mu_0,4,2\mu_0,4,3} F_{\mu_1,5,1\mu_2,5,1\mu_2,5,2}^{\mu_1,5,1} \\
& F_{\mu_1,6,1\mu_1,6,2\mu_1,6,3\mu_3,6,1}^{\mu_1,6,1} F_{\mu_0,7,1\mu_0,7,2\mu_3,7,1\mu_3,7,2\mu_5,7,1}^{\mu_0,7,1} \\
& F_{\mu_2,8,1\mu_2,8,2\mu_3,8,1\mu_4,8,1\mu_5,8,1} F_{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_4,9,1\mu_6,9,1}^{\mu_1,9,1}
\end{aligned}$$

I10_4

$$\begin{aligned}
& F_{\mu_0,4,1\mu_0,4,2\mu_0,6,1\mu_0,7,1\mu_0,7,2} F_{\mu_1,5,1\mu_1,5,2\mu_1,5,3\mu_1,7,1\mu_1,9,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,6,4\mu_2,8,1} F_{\mu_3,7,1\mu_3,7,2\mu_3,8,1\mu_3,9,1\mu_3,9,2}^{\mu_3,7,1} \\
& F_{\mu_4,8,1\mu_4,8,2\mu_4,9,1}^{\mu_0,4,1\mu_0,4,2} F_{\mu_1,5,1\mu_1,5,2\mu_1,5,3}^{\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,6,4} F_{\mu_0,7,1\mu_0,7,2\mu_1,7,1\mu_3,7,1\mu_3,7,2}^{\mu_0,7,1} \\
& F_{\mu_2,8,1\mu_3,8,1\mu_4,8,1\mu_4,8,2\mu_5,8,1} F_{\mu_1,9,1\mu_3,9,1\mu_3,9,2\mu_4,9,1\mu_5,9,1}^{\mu_1,9,1}
\end{aligned}$$

I10_5

$$\begin{aligned}
& F_{\mu_0,4,1\mu_0,4,2\mu_0,6,1\mu_0,7,1\mu_0,9,1} F_{\mu_1,5,1\mu_1,7,1\mu_1,8,1\mu_1,9,1\mu_1,9,2} \\
& F_{\mu_2,5,1\mu_2,5,2\mu_2,5,3\mu_2,7,1\mu_2,7,2} F_{\mu_3,6,1\mu_3,6,2\mu_3,6,3\mu_3,8,1\mu_3,8,2}^{\mu_3,6,1} \\
& F_{\mu_4,6,1\mu_4,8,1\mu_4,9,1}^{\mu_0,4,1\mu_0,4,2} F_{\mu_1,5,1\mu_2,5,1\mu_2,5,2\mu_2,5,3}^{\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_3,6,1\mu_3,6,2\mu_3,6,3\mu_4,6,1} F_{\mu_0,7,1\mu_1,7,1\mu_2,7,1\mu_2,7,2}^{\mu_0,7,1} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_4,8,1\mu_5,8,1} F_{\mu_0,9,1\mu_1,9,1\mu_1,9,2\mu_4,9,1\mu_7,9,1}^{\mu_0,9,1}
\end{aligned}$$

I10_6

$$\begin{aligned}
& F_{\mu_0,4,1\mu_0,4,2\mu_0,6,1\mu_0,8,1\mu_0,9,1} F_{\mu_1,5,1\mu_1,6,1\mu_1,8,1\mu_1,9,1\mu_1,9,2} \\
& F_{\mu_2,5,1\mu_2,5,2\mu_2,7,1\mu_2,7,2\mu_2,8,1} F_{\mu_3,6,1\mu_3,6,2\mu_3,6,3\mu_3,7,1\mu_3,9,1} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2} F_{\mu_5,7,1\mu_5,9,1}^{\mu_1,5,1\mu_2,5,1\mu_2,5,2} \\
& F_{\mu_0,6,1\mu_1,6,1\mu_3,6,1\mu_3,6,2\mu_3,6,3} F_{\mu_2,7,1\mu_2,7,2\mu_3,7,1\mu_4,7,1\mu_5,7,1} \\
& F_{\mu_0,8,1\mu_1,8,1\mu_2,8,1\mu_4,8,1\mu_4,8,2} F_{\mu_0,9,1\mu_1,9,1\mu_1,9,2\mu_3,9,1\mu_5,9,1}
\end{aligned}$$

I10_7

$$\begin{aligned}
& F_{\mu_0,3,1\mu_0,5,1\mu_0,5,2\mu_0,7,1\mu_0,9,1} F_{\mu_1,4,1\mu_1,4,2\mu_1,6,1\mu_1,8,1\mu_1,8,2} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,6,4\mu_2,7,1} F_{\mu_3,5,1\mu_3,5,2\mu_3,8,1\mu_3,9,1}^{\mu_0,3,1} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,9,1}^{\mu_1,4,1\mu_1,4,2} F_{\mu_5,9,1}^{\mu_0,5,1\mu_0,5,2\mu_3,5,1\mu_3,5,2} \\
& F_{\mu_1,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,6,4} F_{\mu_7,8,1\mu_7,9,1}^{\mu_0,7,1\mu_2,7,1\mu_4,7,1} \\
& F_{\mu_1,8,1\mu_1,8,2\mu_3,8,1\mu_4,8,1\mu_7,8,1} F_{\mu_0,9,1\mu_3,9,1\mu_4,9,1\mu_5,9,1\mu_7,9,1}
\end{aligned}$$

I10_8

$$\begin{aligned}
& F_{\mu_0,4,1\mu_0,4,2\mu_0,7,1\mu_0,9,1\mu_0,9,2} F_{\mu_1,5,1\mu_1,5,2\mu_1,5,3\mu_1,5,4\mu_1,8,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,7,1\mu_2,8,1\mu_2,8,2} F_{\mu_3,6,1\mu_3,6,2\mu_3,7,1\mu_3,7,2\mu_3,7,3} \\
& F_{\mu_4,8,1\mu_4,8,2\mu_4,9,1}^{\mu_0,4,1\mu_0,4,2} F_{\mu_5,9,1}^{\mu_1,5,1\mu_1,5,2\mu_1,5,3\mu_1,5,4} \\
& F_{\mu_6,9,1}^{\mu_2,6,1\mu_2,6,2\mu_3,6,1\mu_3,6,2} F_{\mu_0,7,1\mu_2,7,1\mu_3,7,1\mu_3,7,2\mu_3,7,3}^{\mu_0,7,1\mu_2,7,1\mu_3,7,1\mu_3,7,2\mu_3,7,3} \\
& F_{\mu_1,8,1\mu_2,8,1\mu_2,8,2\mu_4,8,1\mu_4,8,2} F_{\mu_0,9,1\mu_0,9,2\mu_4,9,1\mu_5,9,1\mu_6,9,1}
\end{aligned}$$

I10_9

$$\begin{aligned}
& F_{\mu_0,4,1\mu_0,4,2\mu_0,4,3\mu_0,7,1\mu_0,9,1} F_{\mu_1,5,1\mu_1,5,2\mu_1,6,1\mu_1,6,2\mu_1,6,3} \\
& F_{\mu_2,5,1\mu_2,5,2\mu_2,7,1\mu_2,8,1\mu_2,9,1} F_{\mu_3,6,1\mu_3,7,1\mu_3,7,2\mu_3,8,1\mu_3,8,2} \\
& F_{\mu_4,8,1\mu_4,9,1}^{\mu_0,4,1\mu_0,4,2\mu_0,4,3} F_{\mu_5,9,1}^{\mu_1,5,1\mu_1,5,2\mu_2,5,1\mu_2,5,2} \\
& F_{\mu_6,9,1}^{\mu_1,6,1\mu_1,6,2\mu_1,6,3\mu_3,6,1} F_{\mu_7,8,1}^{\mu_0,7,1\mu_2,7,1\mu_3,7,1\mu_3,7,2} \\
& F_{\mu_2,8,1\mu_3,8,1\mu_3,8,2\mu_4,8,1\mu_7,8,1} F_{\mu_0,9,1\mu_2,9,1\mu_4,9,1\mu_5,9,1\mu_6,9,1}
\end{aligned}$$

I10_10

$$\begin{aligned}
& F_{\mu_0,4,1\mu_0,4,2\mu_0,5,1\mu_0,5,2\mu_0,8,1} F_{\mu_1,5,1\mu_1,5,2\mu_1,5,3\mu_1,7,1\mu_1,9,1} \\
& F_{\mu_2,6,1\mu_2,7,1\mu_2,8,1\mu_2,8,2\mu_2,9,1} F_{\mu_3,6,1\mu_3,6,2\mu_3,6,3\mu_3,8,1\mu_3,9,1} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,9,1}^{\mu_0,4,1\mu_0,4,2} F_{\mu_5,9,1}^{\mu_0,5,1\mu_0,5,2\mu_1,5,1\mu_1,5,2\mu_1,5,3} \\
& F_{\mu_6,8,1}^{\mu_2,6,1\mu_3,6,1\mu_3,6,2\mu_3,6,3} F_{\mu_7,9,1}^{\mu_1,7,1\mu_2,7,1\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_0,8,1\mu_2,8,1\mu_2,8,2\mu_3,8,1\mu_6,8,1} F_{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_4,9,1\mu_7,9,1}
\end{aligned}$$

I10_11

$$\begin{aligned}
& F_{\mu_0,3,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,9,1} F_{\mu_1,4,1\mu_1,6,1\mu_1,6,2\mu_1,7,1\mu_1,7,2} \\
& F_{\mu_2,5,1\mu_2,5,2\mu_2,5,3\mu_2,5,4\mu_2,8,1} F_{\mu_3,6,1\mu_3,8,1\mu_3,8,2\mu_3,9,1}^{\mu_0,3,1} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,9,1}^{\mu_1,4,1} F_{\mu_5,9,1}^{\mu_2,5,1\mu_2,5,2\mu_2,5,3\mu_2,5,4} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_1,6,1\mu_1,6,2\mu_3,6,1} F_{\mu_0,7,1\mu_1,7,1\mu_1,7,2\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_8,9,1}^{\mu_2,8,1\mu_3,8,1\mu_3,8,2\mu_4,8,1} F_{\mu_0,9,1\mu_3,9,1\mu_4,9,1\mu_5,9,1\mu_8,9,1}
\end{aligned}$$

I10_12

$$\begin{aligned}
& F_{\mu_0,3,1\mu_0,5,1\mu_0,5,2\mu_0,8,1\mu_0,9,1} F_{\mu_1,4,1\mu_1,4,2\mu_1,6,1\mu_1,7,1\mu_1,9,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,7,1\mu_2,8,1\mu_2,8,2} F_{\mu_3,5,1\mu_3,5,2\mu_3,8,1\mu_3,9,1}^{\mu_0,3,1} \\
& F_{\mu_4,6,1\mu_4,7,1\mu_4,7,2}^{\mu_1,4,1\mu_1,4,2} F_{\mu_5,8,1}^{\mu_0,5,1\mu_0,5,2\mu_3,5,1\mu_3,5,2} \\
& F_{\mu_6,9,1}^{\mu_1,6,1\mu_2,6,1\mu_2,6,2\mu_4,6,1} F_{\mu_7,9,1}^{\mu_1,7,1\mu_2,7,1\mu_4,7,1\mu_4,7,2} \\
& F^{\mu_0,8,1\mu_2,8,1\mu_2,8,2\mu_3,8,1\mu_5,8,1} F^{\mu_0,9,1\mu_1,9,1\mu_3,9,1\mu_6,9,1\mu_7,9,1}
\end{aligned}$$

I12_1

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,7,2\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,9,2\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,9,1\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,8,3\mu_3,9,1} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,10,1\mu_4,11,1\mu_4,11,2} F_{\mu_5,6,1\mu_5,10,1\mu_5,10,2}^{\mu_0,5,1\mu_1,5,1} \\
& F^{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_0,7,2\mu_3,7,1\mu_4,7,1} \\
& F^{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_3,8,3\mu_4,8,1} F_{\mu_9,11,1}^{\mu_1,9,1\mu_1,9,2\mu_2,9,1\mu_3,9,1} \\
& F^{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_5,10,1\mu_5,10,2} F^{\mu_1,11,1\mu_4,11,1\mu_4,11,2\mu_7,11,1\mu_9,11,1}
\end{aligned}$$

I12_2

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,11,1\mu_1,11,2} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,9,2\mu_2,10,1\mu_2,10,2} F_{\mu_3,7,1\mu_3,7,2\mu_3,8,1\mu_3,9,1\mu_3,11,1} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2\mu_4,8,3\mu_4,10,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F^{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_3,7,2\mu_4,7,1} \\
& F^{\mu_1,8,1\mu_3,8,1\mu_4,8,1\mu_4,8,2\mu_4,8,3} F_{\mu_9,11,1}^{\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_3,9,1} \\
& F^{\mu_0,10,1\mu_2,10,1\mu_2,10,2\mu_4,10,1\mu_5,10,1} F^{\mu_1,11,1\mu_1,11,2\mu_3,11,1\mu_7,11,1\mu_9,11,1}
\end{aligned}$$

I12_3

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,10,1\mu_0,10,2} F_{\mu_1,5,1\mu_1,8,1\mu_1,8,2\mu_1,9,1\mu_1,9,2} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,10,1\mu_2,11,1\mu_2,11,2} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_6,11,1}^{\mu_0,6,1\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F^{\mu_1,8,1\mu_1,8,2\mu_3,8,1\mu_3,8,2\mu_4,8,1} F^{\mu_1,9,1\mu_1,9,2\mu_2,9,1\mu_3,9,1\mu_3,9,2} \\
& F^{\mu_0,10,1\mu_0,10,2\mu_2,10,1\mu_4,10,1\mu_5,10,1} F^{\mu_2,11,1\mu_2,11,2\mu_4,11,1\mu_6,11,1\mu_7,11,1}
\end{aligned}$$

I12_4

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,10,1\mu_0,10,2} F_{\mu_1,5,1\mu_1,5,2\mu_1,8,1\mu_1,9,1\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,9,1\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,9,2\mu_3,9,3} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1\mu_1,5,2} \\
& F^{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_8,11,1\mu_8,11,2}^{\mu_1,8,1\mu_3,8,1\mu_4,8,1} F^{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_3,9,2\mu_3,9,3} \\
& F^{\mu_0,10,1\mu_0,10,2\mu_2,10,1\mu_4,10,1\mu_5,10,1} F^{\mu_1,11,1\mu_4,11,1\mu_7,11,1\mu_8,11,1\mu_8,11,2}
\end{aligned}$$

I12_5

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,11,1\mu_1,11,2} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,9,2\mu_2,10,1\mu_2,10,2} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,8,3\mu_3,9,1} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_3,8,3\mu_4,8,1} F_{\mu_9,11,1}^{\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_2,10,2\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_1,11,2\mu_4,11,1\mu_7,11,1\mu_9,11,1}
\end{aligned}$$

I12_6

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,8,2\mu_1,9,1\mu_1,9,2} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,10,1\mu_2,11,1\mu_2,11,2} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,10,1\mu_5,10,2}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_6,11,1}^{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_1,8,1\mu_1,8,2\mu_3,8,1\mu_3,8,2\mu_4,8,1} F_{\mu_1,9,1\mu_1,9,2\mu_2,9,1\mu_3,9,1\mu_3,9,2}^{\mu_1,9,1\mu_1,9,2\mu_2,9,1\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_5,10,1\mu_5,10,2} F_{\mu_2,11,1\mu_2,11,2\mu_4,11,1\mu_6,11,1\mu_7,11,1}
\end{aligned}$$

I12_7

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,7,2\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,11,1\mu_1,11,2} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,9,1\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,9,2\mu_3,9,3} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,10,1\mu_5,10,2}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_0,7,2\mu_3,7,1\mu_4,7,1} \\
& F_{\mu_8,11,1}^{\mu_1,8,1\mu_3,8,1\mu_4,8,1\mu_4,8,2} F_{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_3,9,2\mu_3,9,3}^{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_3,9,2\mu_3,9,3} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_5,10,1\mu_5,10,2} F_{\mu_1,11,1\mu_1,11,2\mu_4,11,1\mu_7,11,1\mu_8,11,1}
\end{aligned}$$

I12_8

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,9,2\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,10,1\mu_2,10,2\mu_2,11,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,9,2\mu_3,11,1} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,7,3\mu_4,8,1\mu_4,10,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2\mu_4,7,3} \\
& F_{\mu_8,11,1}^{\mu_1,8,1\mu_3,8,1\mu_4,8,1} F_{\mu_1,9,1\mu_1,9,2\mu_2,9,1\mu_3,9,1\mu_3,9,2}^{\mu_1,9,1\mu_1,9,2\mu_2,9,1\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_2,10,2\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_2,11,1\mu_3,11,1\mu_8,11,1\mu_8,11,2}
\end{aligned}$$

I12_9

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,9,2\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,9,2\mu_2,10,1\mu_2,10,2} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,11,1\mu_3,11,2} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,8,2\mu_4,10,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_8,11,1}^{\mu_1,8,1\mu_3,8,1\mu_4,8,1\mu_4,8,2} F_{\mu_1,9,1\mu_1,9,2\mu_2,9,1\mu_2,9,2\mu_3,9,1}^{\mu_1,9,1\mu_1,9,2\mu_2,9,1\mu_2,9,2\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_2,10,2\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_3,11,1\mu_3,11,2\mu_7,11,1\mu_8,11,1}
\end{aligned}$$

I12_10

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,10,1\mu_0,10,2} F_{\mu_1,5,1\mu_1,5,2\mu_1,8,1\mu_1,9,1\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,9,1\mu_2,10,1} F_{\mu_3,7,1\mu_3,7,2\mu_3,8,1\mu_3,9,1\mu_3,11,1} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2\mu_4,8,3\mu_4,10,1} F_{\mu_5,6,1\mu_5,5,1\mu_5,10,1} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_3,7,2\mu_4,7,1} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_4,8,1\mu_4,8,2\mu_4,8,3} F_{\mu_9,11,1}^{\mu_1,9,1\mu_2,9,1\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_0,10,2\mu_2,10,1\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1}^{\mu_1,11,1\mu_3,11,1\mu_7,11,1\mu_9,11,1\mu_9,11,2}
\end{aligned}$$

I12_11

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,9,2\mu_1,9,3} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,10,1\mu_2,10,2\mu_2,11,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,11,1\mu_3,11,2} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,8,2\mu_4,10,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_8,11,1}^{\mu_1,8,1\mu_3,8,1\mu_4,8,1\mu_4,8,2} F_{\mu_1,9,1\mu_1,9,2\mu_1,9,3\mu_2,9,1\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_2,10,2\mu_4,10,1\mu_5,10,1} F_{\mu_2,11,1}^{\mu_2,11,1\mu_3,11,1\mu_3,11,2\mu_7,11,1\mu_8,11,1}
\end{aligned}$$

I12_12

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,11,1\mu_1,11,2} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,9,2\mu_2,10,1\mu_2,10,2} F_{\mu_3,7,1\mu_3,7,2\mu_3,8,1\mu_3,8,2\mu_3,9,1} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_3,7,2\mu_4,7,1} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_4,8,1\mu_4,8,2} F_{\mu_9,11,1}^{\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_2,10,2\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1}^{\mu_1,11,2\mu_4,11,1\mu_7,11,1\mu_9,11,1}
\end{aligned}$$

I12_13

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,5,2\mu_0,6,1\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,9,2\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,9,1\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,8,3\mu_3,9,1} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,10,1\mu_4,10,2\mu_4,11,1} F_{\mu_5,6,1\mu_5,5,1\mu_5,10,1}^{\mu_0,5,1\mu_0,5,2\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_3,8,3\mu_4,8,1} F_{\mu_9,11,1}^{\mu_1,9,1\mu_1,9,2\mu_2,9,1\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_4,10,2\mu_5,10,1} F_{\mu_1,11,1}^{\mu_1,11,1\mu_4,11,1\mu_7,11,1\mu_7,11,2\mu_9,11,1}
\end{aligned}$$

I12_14

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,9,2\mu_1,9,3} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,10,1\mu_2,10,2\mu_2,11,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,9,1\mu_3,11,1} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_4,8,1\mu_4,8,2} F_{\mu_1,9,1\mu_1,9,2\mu_1,9,3\mu_2,9,1\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_2,10,2\mu_4,10,1\mu_5,10,1} F_{\mu_2,11,1}^{\mu_2,11,1\mu_3,11,1\mu_4,11,1\mu_7,11,1\mu_7,11,2}
\end{aligned}$$

I12_15

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,7,2\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,9,2\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,9,1\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,11,1\mu_3,11,2} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2\mu_4,8,3\mu_4,10,1} F_{\mu_5,6,1\mu_5,10,1\mu_5,10,2}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_0,7,2\mu_3,7,1\mu_4,7,1} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_4,8,1\mu_4,8,2\mu_4,8,3} F_{\mu_9,11,1}^{\mu_1,9,1\mu_1,9,2\mu_2,9,1\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_5,10,1\mu_5,10,2} F_{\mu_1,11,1\mu_3,11,1\mu_3,11,2\mu_7,11,1\mu_9,11,1}
\end{aligned}$$

I12_16

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,11,1\mu_1,11,2} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,9,2\mu_2,10,1\mu_2,10,2} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,9,1\mu_3,11,1} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,8,2\mu_4,10,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_4,8,1\mu_4,8,2} F_{\mu_9,11,1}^{\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_2,10,2\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_1,11,2\mu_3,11,1\mu_7,11,1\mu_9,11,1}
\end{aligned}$$

I12_17

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,10,1\mu_0,10,2} F_{\mu_1,5,1\mu_1,5,2\mu_1,8,1\mu_1,9,1\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,9,1\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,9,2\mu_3,9,3} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1\mu_1,5,2} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_5,6,1} F_{\mu_7,11,1\mu_7,11,2}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1} \\
& F_{\mu_8,11,1}^{\mu_1,8,1\mu_3,8,1\mu_4,8,1\mu_4,8,2} F_{\mu_3,9,1\mu_3,9,2\mu_3,9,3}^{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_3,9,2\mu_3,9,3} \\
& F_{\mu_0,10,1\mu_0,10,2\mu_2,10,1\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_4,11,1\mu_7,11,1\mu_7,11,2\mu_8,11,1}
\end{aligned}$$

I12_18

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,5,2\mu_0,6,1\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,8,2\mu_1,9,1\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,9,1\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,9,1\mu_3,11,1} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,10,1\mu_4,10,2} F_{\mu_5,6,1\mu_5,10,1}^{\mu_0,5,1\mu_0,5,2\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_1,8,1\mu_1,8,2\mu_3,8,1\mu_3,8,2\mu_4,8,1} F_{\mu_9,11,1\mu_9,11,2}^{\mu_1,9,1\mu_2,9,1\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_4,10,2\mu_5,10,1} F_{\mu_1,11,1\mu_3,11,1\mu_7,11,1\mu_9,11,1\mu_9,11,2}
\end{aligned}$$

I12_19

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,11,1\mu_1,11,2} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,9,2\mu_2,10,1\mu_2,10,2} F_{\mu_3,7,1\mu_3,7,2\mu_3,8,1\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_3,7,2\mu_4,7,1} \\
& F_{\mu_8,11,1}^{\mu_1,8,1\mu_3,8,1\mu_4,8,1\mu_4,8,2} F_{\mu_3,9,1\mu_3,9,2\mu_3,9,1\mu_3,9,2}^{\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_2,10,2\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_1,11,2\mu_4,11,1\mu_7,11,1\mu_8,11,1}
\end{aligned}$$

I12_20

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,10,1\mu_0,10,2} F_{\mu_1,5,1\mu_1,5,2\mu_1,8,1\mu_1,9,1\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,9,1\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,9,2\mu_3,11,1} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2\mu_4,8,3\mu_4,10,1} F_{\mu_5,6,1\mu_5,5,1\mu_5,10,1} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_5,6,1} F_{\mu_7,11,1\mu_7,11,2} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_4,8,1\mu_4,8,2\mu_4,8,3} F_{\mu_9,11,1\mu_9,11,2} \\
& F_{\mu_0,10,1\mu_0,10,2\mu_2,10,1\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_3,11,1\mu_7,11,1\mu_7,11,2\mu_9,11,1}
\end{aligned}$$

I12_21

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,5,2\mu_0,6,1\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,8,2\mu_1,9,1\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,9,1\mu_2,10,1} F_{\mu_3,7,1\mu_3,7,2\mu_3,8,1\mu_3,9,1\mu_3,11,1} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2\mu_4,10,1\mu_4,10,2} F_{\mu_5,6,1\mu_5,5,1\mu_5,10,1} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_5,6,1} F_{\mu_7,11,1\mu_7,11,2} \\
& F_{\mu_1,8,1\mu_1,8,2\mu_3,8,1\mu_4,8,1\mu_4,8,2} F_{\mu_9,11,1\mu_9,11,2} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_4,10,2\mu_5,10,1} F_{\mu_1,11,1\mu_3,11,1\mu_7,11,1\mu_9,11,1\mu_9,11,2}
\end{aligned}$$

I12_22

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,9,2\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,9,2\mu_2,10,1\mu_2,10,2} F_{\mu_3,7,1\mu_3,7,2\mu_3,8,1\mu_3,8,2\mu_3,9,1} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,10,1\mu_4,11,1\mu_4,11,2} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1\mu_7,11,2} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_4,8,1} F_{\mu_8,11,1\mu_8,11,2} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_2,10,2\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_4,11,1\mu_4,11,2\mu_7,11,1\mu_8,11,1}
\end{aligned}$$

I12_23

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,5,2\mu_0,6,1\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,8,2\mu_1,9,1\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,9,1\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,9,2\mu_3,9,3} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,10,1\mu_4,10,2\mu_4,11,1} F_{\mu_5,6,1\mu_5,5,1\mu_5,10,1} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_5,6,1} F_{\mu_7,11,1\mu_7,11,2} \\
& F_{\mu_1,8,1\mu_1,8,2\mu_3,8,1\mu_4,8,1} F_{\mu_8,11,1\mu_8,11,2} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_4,10,2\mu_5,10,1} F_{\mu_1,11,1\mu_4,11,1\mu_4,11,2\mu_7,11,1\mu_7,11,2\mu_8,11,1}
\end{aligned}$$

I12_24

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,7,2\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,11,1\mu_1,11,2} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,9,1\mu_2,10,1\mu_2,10,2} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,9,2\mu_3,9,3} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1\mu_7,11,2} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_4,8,1\mu_4,8,2} F_{\mu_8,11,1\mu_8,11,2} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_2,10,2\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_1,11,2\mu_4,11,1\mu_7,11,1\mu_8,11,1}
\end{aligned}$$

I12_25

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,8,2\mu_1,9,1\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,9,2\mu_2,10,1\mu_2,10,2} F_{\mu_3,7,1\mu_3,7,2\mu_3,8,1\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,10,1\mu_4,11,1\mu_4,11,2} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_3,7,2\mu_4,7,1} \\
& F_{\mu_8,11,1}^{\mu_1,8,1\mu_1,8,2\mu_3,8,1\mu_4,8,1} F_{\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_2,10,2\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_4,11,1\mu_4,11,2\mu_7,11,1\mu_8,11,1}
\end{aligned}$$

I12_26

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,7,2\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,11,1\mu_1,11,2} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,9,1\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,9,2\mu_3,11,1} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2\mu_4,8,3\mu_4,10,1} F_{\mu_5,6,1\mu_5,10,1\mu_5,10,2}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_0,7,2\mu_3,7,1\mu_4,7,1} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_4,8,1\mu_4,8,2\mu_4,8,3} F_{\mu_9,11,1}^{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_5,10,1\mu_5,10,2} F_{\mu_1,11,1\mu_1,11,2\mu_3,11,1\mu_7,11,1\mu_9,11,1}
\end{aligned}$$

I12_27

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,10,1\mu_0,10,2} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,11,1\mu_1,11,2} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,9,1\mu_2,9,2\mu_2,10,1} F_{\mu_3,7,1\mu_3,7,2\mu_3,8,1\mu_3,9,1\mu_3,11,1} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2\mu_4,8,3\mu_4,10,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_3,7,2\mu_4,7,1} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_4,8,1\mu_4,8,2\mu_4,8,3} F_{\mu_9,11,1}^{\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_0,10,2\mu_2,10,1\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_1,11,2\mu_3,11,1\mu_7,11,1\mu_9,11,1}
\end{aligned}$$

I12_28

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,10,1\mu_0,10,2} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,11,1\mu_1,11,2} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,9,1\mu_2,9,2\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,9,1\mu_3,11,1} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,8,2\mu_4,10,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_4,8,1\mu_4,8,2} F_{\mu_9,11,1}^{\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_0,10,2\mu_2,10,1\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_1,11,2\mu_3,11,1\mu_7,11,1\mu_9,11,1}
\end{aligned}$$

I12_29

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,10,1\mu_0,10,2} F_{\mu_1,5,1\mu_1,5,2\mu_1,8,1\mu_1,9,1\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,9,1\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,8,3\mu_3,9,1} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1\mu_1,5,2} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_3,8,3\mu_4,8,1} F_{\mu_9,11,1\mu_9,11,2}^{\mu_1,9,1\mu_2,9,1\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_0,10,2\mu_2,10,1\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_4,11,1\mu_7,11,1\mu_9,11,1\mu_9,11,2}
\end{aligned}$$

I12_30

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,8,2\mu_1,8,3\mu_1,9,1} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,9,2\mu_2,10,1\mu_2,11,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,11,1\mu_3,11,2} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,10,1\mu_4,10,2} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_1,8,1\mu_1,8,2\mu_1,8,3\mu_3,8,1\mu_4,8,1} F_{\mu_9,11,1}^{\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_4,10,2\mu_5,10,1} F_{\mu_2,11,1\mu_3,11,1\mu_3,11,2\mu_7,11,1\mu_9,11,1}
\end{aligned}$$

I12_31

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,8,2\mu_1,9,1\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,9,1\mu_2,9,2\mu_2,10,1} F_{\mu_3,7,1\mu_3,7,2\mu_3,8,1\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,10,1\mu_4,11,1\mu_4,11,2} F_{\mu_5,6,1\mu_5,10,1\mu_5,10,2}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_2,6,2\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_3,7,2\mu_4,7,1} \\
& F_{\mu_8,11,1}^{\mu_1,8,1\mu_1,8,2\mu_3,8,1\mu_4,8,1} F_{\mu_9,11,1}^{\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_5,10,1\mu_5,10,2} F_{\mu_1,11,1\mu_4,11,1\mu_4,11,2\mu_7,11,1\mu_8,11,1}
\end{aligned}$$

I12_32

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,7,2\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,8,2\mu_1,9,1\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,9,1\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,9,2\mu_3,9,3} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,10,1\mu_4,11,1\mu_4,11,2} F_{\mu_5,6,1\mu_5,10,1\mu_5,10,2}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_0,7,2\mu_3,7,1\mu_4,7,1} \\
& F_{\mu_8,11,1}^{\mu_1,8,1\mu_1,8,2\mu_3,8,1\mu_4,8,1} F_{\mu_9,11,1}^{\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_5,10,1\mu_5,10,2} F_{\mu_1,11,1\mu_4,11,1\mu_4,11,2\mu_7,11,1\mu_8,11,1}
\end{aligned}$$

I12_33

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,7,2\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,9,2\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,9,1\mu_2,10,1\mu_2,10,2} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,11,1\mu_3,11,2} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2\mu_4,8,3\mu_4,10,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_0,7,2\mu_3,7,1\mu_4,7,1} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_4,8,1\mu_4,8,2\mu_4,8,3} F_{\mu_9,11,1}^{\mu_1,9,1\mu_1,9,2\mu_2,9,1\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_2,10,2\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_3,11,1\mu_3,11,2\mu_7,11,1\mu_9,11,1}
\end{aligned}$$

I12_34

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,8,2\mu_1,9,1\mu_1,9,2} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,10,1\mu_2,10,2\mu_2,11,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,9,2\mu_3,11,1} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1\mu_7,11,2}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1} \\
& F_{\mu_1,8,1\mu_1,8,2\mu_3,8,1\mu_4,8,1\mu_4,8,2} F_{\mu_9,11,1\mu_9,11,2}^{\mu_1,9,1\mu_1,9,2\mu_2,9,1\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_2,10,2\mu_4,10,1\mu_5,10,1} F_{\mu_2,11,1\mu_3,11,1\mu_4,11,1\mu_7,11,1\mu_7,11,2}
\end{aligned}$$

I12_35

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,9,2\mu_1,9,3} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,10,1\mu_2,11,1\mu_2,11,2} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,8,3\mu_3,9,1} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,10,1\mu_5,10,2} \\
& F_{\mu_6,11,1}^{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F^{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_3,8,3\mu_4,8,1} F^{\mu_1,9,1\mu_1,9,2\mu_1,9,3\mu_2,9,1\mu_3,9,1} \\
& F^{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_5,10,1\mu_5,10,2} F^{\mu_2,11,1\mu_2,11,2\mu_4,11,1\mu_6,11,1\mu_7,11,1}
\end{aligned}$$

I12_36

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,11,1\mu_1,11,2} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,9,1\mu_2,9,2\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,8,3\mu_3,9,1} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,10,1\mu_5,10,2}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_6,11,1}^{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_2,6,2\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F^{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_3,8,3\mu_4,8,1} F^{\mu_1,9,1\mu_1,9,2\mu_1,9,3\mu_2,9,1\mu_3,9,1} \\
& F^{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_5,10,1\mu_5,10,2} F^{\mu_1,11,1\mu_1,11,2\mu_4,11,1\mu_7,11,1\mu_9,11,1}
\end{aligned}$$

I12_37

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,10,1\mu_0,10,2} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,9,2\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,9,1\mu_2,9,2\mu_2,10,1} F_{\mu_3,7,1\mu_3,7,2\mu_3,8,1\mu_3,8,2\mu_3,9,1} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,10,1\mu_4,11,1\mu_4,11,2} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_6,11,1}^{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_3,7,2\mu_4,7,1} \\
& F_{\mu_8,11,1}^{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_4,8,1} F^{\mu_1,9,1\mu_1,9,2\mu_2,9,1\mu_2,9,2\mu_3,9,1} \\
& F^{\mu_0,10,1\mu_0,10,2\mu_2,10,1\mu_4,10,1\mu_5,10,1} F^{\mu_1,11,1\mu_4,11,1\mu_4,11,2\mu_7,11,1\mu_8,11,1}
\end{aligned}$$

I12_38

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,10,1\mu_0,10,2} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,9,2\mu_1,9,3} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,10,1\mu_2,11,1\mu_2,11,2} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,8,3\mu_3,9,1} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_6,11,1}^{\mu_0,6,1\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F^{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_3,8,3\mu_4,8,1} F^{\mu_1,9,1\mu_1,9,2\mu_1,9,3\mu_2,9,1\mu_3,9,1} \\
& F^{\mu_0,10,1\mu_0,10,2\mu_2,10,1\mu_4,10,1\mu_5,10,1} F^{\mu_2,11,1\mu_2,11,2\mu_4,11,1\mu_6,11,1\mu_7,11,1}
\end{aligned}$$

I12_39

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,10,1\mu_0,10,2} F_{\mu_1,5,1\mu_1,5,2\mu_1,8,1\mu_1,9,1\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,9,1\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,9,1\mu_3,11,1} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,8,2\mu_4,10,1} F_{\mu_5,6,1\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1\mu_1,5,2} \\
& F_{\mu_6,11,1}^{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F^{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_4,8,1\mu_4,8,2} F^{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_9,11,1\mu_9,11,2} \\
& F^{\mu_0,10,1\mu_0,10,2\mu_2,10,1\mu_4,10,1\mu_5,10,1} F^{\mu_1,11,1\mu_3,11,1\mu_7,11,1\mu_9,11,1\mu_9,11,2}
\end{aligned}$$

I12_40

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,7,2\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,8,2\mu_1,9,1\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,9,1\mu_2,10,1\mu_2,10,2} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,10,1\mu_4,11,1\mu_4,11,2} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_0,7,2\mu_3,7,1\mu_4,7,1} \\
& F_{\mu_1,8,1\mu_1,8,2\mu_3,8,1\mu_3,8,2\mu_4,8,1} F_{\mu_9,11,1}^{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_2,10,2\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_4,11,1\mu_4,11,2\mu_7,11,1\mu_9,11,1}
\end{aligned}$$

I12_41

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,10,1\mu_0,10,2} F_{\mu_1,5,1\mu_1,8,1\mu_1,8,2\mu_1,8,3\mu_1,9,1} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,10,1\mu_2,11,1\mu_2,11,2} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,9,2\mu_3,9,3} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_6,11,1}^{\mu_0,6,1\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_1,8,1\mu_1,8,2\mu_1,8,3\mu_3,8,1\mu_4,8,1} F_{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_3,9,2\mu_3,9,3}^{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_3,9,2\mu_3,9,3} \\
& F_{\mu_0,10,1\mu_0,10,2\mu_2,10,1\mu_4,10,1\mu_5,10,1} F_{\mu_2,11,1\mu_2,11,2\mu_4,11,1\mu_6,11,1\mu_7,11,1}
\end{aligned}$$

I12_42

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,5,2\mu_1,8,1\mu_1,9,1\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,9,1\mu_2,10,1\mu_2,10,2} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,8,3\mu_3,9,1} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1\mu_1,5,2} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_2,6,2\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_3,8,3\mu_4,8,1} F_{\mu_9,11,1\mu_9,11,2}^{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_2,10,2\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_4,11,1\mu_7,11,1\mu_9,11,1\mu_9,11,2}
\end{aligned}$$

I12_43

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,11,1\mu_1,11,2} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,10,1\mu_2,10,2\mu_2,11,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,8,2\mu_4,10,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_4,8,1\mu_4,8,2} F_{\mu_9,11,1}^{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_2,10,2\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_1,11,2\mu_2,11,1\mu_7,11,1\mu_9,11,1}
\end{aligned}$$

I12_44

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,8,2\mu_1,8,3\mu_1,9,1} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,10,1\mu_2,11,1\mu_2,11,2} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,9,2\mu_3,9,3} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,2}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_6,11,1}^{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_1,8,1\mu_1,8,2\mu_1,8,3\mu_3,8,1\mu_4,8,1} F_{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_3,9,2\mu_3,9,3}^{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_3,9,2\mu_3,9,3} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_5,10,1\mu_5,10,2} F_{\mu_2,11,1\mu_2,11,2\mu_4,11,1\mu_6,11,1\mu_7,11,1}
\end{aligned}$$

I12_45

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,7,2\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,8,2\mu_1,9,1\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,9,1\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,10,1\mu_4,11,1\mu_4,11,2} F_{\mu_5,6,1\mu_5,10,1\mu_5,10,2}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_5,6,1}^{\mu_0,7,1\mu_0,7,2\mu_3,7,1\mu_4,7,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_0,7,2\mu_3,7,1\mu_4,7,1} \\
& F_{\mu_1,8,1\mu_1,8,2\mu_3,8,1\mu_3,8,2\mu_4,8,1} F_{\mu_9,11,1}^{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_5,10,1\mu_5,10,2} F_{\mu_1,11,1\mu_4,11,1\mu_4,11,2\mu_7,11,1\mu_9,11,1}^{\mu_1,11,1\mu_4,11,1\mu_4,11,2\mu_7,11,1\mu_9,11,1}
\end{aligned}$$

I12_46

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,11,1\mu_1,11,2} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,9,2\mu_2,10,1\mu_2,11,1} F_{\mu_3,7,1\mu_3,7,2\mu_3,8,1\mu_3,8,2\mu_3,9,1} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2\mu_4,10,1\mu_4,10,2} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_3,7,2\mu_4,7,1} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_4,8,1\mu_4,8,2} F_{\mu_9,11,1}^{\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_4,10,2\mu_5,10,1} F_{\mu_1,11,1\mu_1,11,2\mu_2,11,1\mu_7,11,1\mu_9,11,1}^{\mu_1,11,1\mu_1,11,2\mu_2,11,1\mu_7,11,1\mu_9,11,1}
\end{aligned}$$

I12_47

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,8,2\mu_1,8,3\mu_1,9,1} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,9,2\mu_2,9,3\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,11,1\mu_3,11,2} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,10,1\mu_5,10,2}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1} F_{\mu_6,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_1,8,1\mu_1,8,2\mu_1,8,3\mu_3,8,1\mu_4,8,1} F_{\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_2,9,3\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_5,10,1\mu_5,10,2} F_{\mu_3,11,1\mu_3,11,2\mu_4,11,1\mu_6,11,1\mu_7,11,1}^{\mu_3,11,1\mu_3,11,2\mu_4,11,1\mu_6,11,1\mu_7,11,1}
\end{aligned}$$

I12_48

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,5,2\mu_0,6,1\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,9,2\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,9,1\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,9,2\mu_3,11,1} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,10,1\mu_4,10,2} F_{\mu_5,6,1\mu_5,10,1}^{\mu_0,5,1\mu_0,5,2\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_4,8,1} F_{\mu_8,11,1}^{\mu_1,9,1\mu_1,9,2\mu_2,9,1\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_4,10,2\mu_5,10,1} F_{\mu_1,11,1\mu_3,11,1\mu_7,11,1\mu_8,11,1\mu_8,11,2}^{\mu_1,11,1\mu_3,11,1\mu_7,11,1\mu_8,11,1\mu_8,11,2}
\end{aligned}$$

I12_49

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,9,2\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,9,1\mu_2,9,2\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,11,1\mu_3,11,2} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,8,2\mu_4,10,1} F_{\mu_5,6,1\mu_5,10,1\mu_5,10,2}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_2,6,2\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_4,8,1\mu_4,8,2} F_{\mu_8,11,1}^{\mu_1,9,1\mu_1,9,2\mu_2,9,1\mu_2,9,2\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_5,10,1\mu_5,10,2} F_{\mu_1,11,1\mu_3,11,1\mu_3,11,2\mu_7,11,1\mu_8,11,1}^{\mu_1,11,1\mu_3,11,1\mu_3,11,2\mu_7,11,1\mu_8,11,1}
\end{aligned}$$

I12_50

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,10,1\mu_0,10,2} F_{\mu_1,5,1\mu_1,8,1\mu_1,8,2\mu_1,8,3\mu_1,9,1} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,9,2\mu_2,9,3\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,11,1\mu_3,11,2} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,8,1\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{F_{\mu_0,5,1\mu_1,5,1}} \\
& F_{\mu_6,11,1}^{\mu_0,6,1\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2} \\
& F_{\mu_1,8,1\mu_1,8,2\mu_1,8,3\mu_3,8,1\mu_4,8,1} F_{\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_2,9,3\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_0,10,2\mu_2,10,1\mu_4,10,1\mu_5,10,1} F_{\mu_3,11,1\mu_3,11,2\mu_4,11,1\mu_6,11,1\mu_7,11,1}
\end{aligned}$$

I12_51

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,11,1\mu_1,11,2} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,9,1\mu_2,9,2\mu_2,10,1} F_{\mu_3,7,1\mu_3,7,2\mu_3,8,1\mu_3,9,1\mu_3,11,1} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2\mu_4,8,3\mu_4,10,1} F_{\mu_5,6,1\mu_5,10,1\mu_5,10,2}^{F_{\mu_0,5,1\mu_1,5,1}} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_2,6,2\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_3,7,2\mu_4,7,1} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_4,8,1\mu_4,8,2\mu_4,8,3} F_{\mu_9,11,1}^{\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_5,10,1\mu_5,10,2} F_{\mu_1,11,1\mu_1,11,2\mu_3,11,1\mu_7,11,1\mu_9,11,1}
\end{aligned}$$

I12_52

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,10,1\mu_0,10,2} F_{\mu_1,5,1\mu_1,8,1\mu_1,8,2\mu_1,9,1\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,9,1\mu_2,9,2\mu_2,10,1} F_{\mu_3,7,1\mu_3,7,2\mu_3,8,1\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,10,1\mu_4,11,1\mu_4,11,2} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{F_{\mu_0,5,1\mu_1,5,1}} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_3,7,2\mu_4,7,1} \\
& F_{\mu_8,11,1}^{\mu_1,8,1\mu_1,8,2\mu_3,8,1\mu_4,8,1} F_{\mu_8,11,1}^{\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_0,10,2\mu_2,10,1\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_4,11,1\mu_4,11,2\mu_7,11,1\mu_8,11,1}
\end{aligned}$$

I12_53

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,11,1\mu_1,11,2} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,9,2\mu_2,9,3\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,9,1\mu_3,11,1} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,7,3\mu_4,8,1\mu_4,10,1} F_{\mu_5,6,1\mu_5,10,1\mu_5,10,2}^{F_{\mu_0,5,1\mu_1,5,1}} \\
& F_{\mu_6,11,1}^{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2\mu_4,7,3} \\
& F_{\mu_8,11,1}^{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_4,8,1} F_{\mu_8,11,1}^{\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_2,9,3\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_5,10,1\mu_5,10,2} F_{\mu_1,11,1\mu_1,11,2\mu_3,11,1\mu_6,11,1\mu_8,11,1}
\end{aligned}$$

I12_54

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,5,2\mu_0,6,1\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,8,2\mu_1,9,1\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_2,9,1\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,10,1\mu_4,10,2\mu_4,11,1} F_{\mu_5,6,1\mu_5,10,1}^{F_{\mu_0,5,1\mu_0,5,2\mu_1,5,1}} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_2,6,3\mu_5,6,1} F_{\mu_7,11,1\mu_7,11,2}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1} \\
& F_{\mu_1,8,1\mu_1,8,2\mu_3,8,1\mu_3,8,2\mu_4,8,1} F_{\mu_9,11,1}^{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_4,10,2\mu_5,10,1} F_{\mu_1,11,1\mu_4,11,1\mu_7,11,1\mu_7,11,2\mu_9,11,1}
\end{aligned}$$

I12_55

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,11,1\mu_1,11,2} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,9,1\mu_2,9,2\mu_2,10,1} F_{\mu_3,7,1\mu_3,7,2\mu_3,8,1\mu_3,8,2\mu_3,9,1} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,10,1\mu_5,10,2} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_2,6,2\mu_5,6,1} F_{\mu_7,11,1}^{\mu_0,7,1\mu_3,7,1\mu_3,7,2\mu_4,7,1} \\
& F_{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_4,8,1\mu_4,8,2} F_{\mu_9,11,1}^{\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_4,10,1\mu_5,10,1\mu_5,10,2} F_{\mu_1,11,1\mu_1,11,2\mu_4,11,1\mu_7,11,1\mu_9,11,1}
\end{aligned}$$

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$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,10,1\mu_0,10,2} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,11,1\mu_1,11,2} \\
& F_{\mu_2,6,1\mu_2,9,1\mu_2,9,2\mu_2,9,3\mu_2,10,1} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,9,1\mu_3,11,1} \\
& F_{\mu_4,7,1\mu_4,7,2\mu_4,7,3\mu_4,8,1\mu_4,10,1} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_6,11,1}^{\mu_0,6,1\mu_2,6,1\mu_5,6,1\mu_5,6,2} F_{\mu_0,7,1\mu_3,7,1\mu_4,7,1\mu_4,7,2\mu_4,7,3} \\
& F_{\mu_8,11,1}^{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_4,8,1} F_{\mu_1,9,1\mu_2,9,1\mu_2,9,2\mu_2,9,3\mu_3,9,1} \\
& F_{\mu_0,10,1\mu_0,10,2\mu_2,10,1\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_1,11,2\mu_3,11,1\mu_6,11,1\mu_8,11,1}
\end{aligned}$$

I12_57

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,6,2\mu_0,7,1\mu_0,10,1} F_{\mu_1,5,1\mu_1,5,2\mu_1,8,1\mu_1,9,1\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,9,1\mu_2,10,1\mu_2,10,2} F_{\mu_3,7,1\mu_3,8,1\mu_3,9,1\mu_3,9,2\mu_3,9,3} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,8,2\mu_4,10,1\mu_4,11,1} F_{\mu_5,6,1\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1\mu_1,5,2} \\
& F_{\mu_0,6,1\mu_0,6,2\mu_2,6,1\mu_2,6,2\mu_5,6,1} F_{\mu_7,11,1\mu_7,11,2}^{\mu_0,7,1\mu_3,7,1\mu_4,7,1} \\
& F_{\mu_8,11,1}^{\mu_1,8,1\mu_3,8,1\mu_4,8,1\mu_4,8,2} F_{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_3,9,2\mu_3,9,3} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_2,10,2\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_4,11,1\mu_7,11,1\mu_7,11,2\mu_8,11,1}
\end{aligned}$$

I12_58

$$\begin{aligned}
& F_{\mu_0,5,1\mu_0,6,1\mu_0,7,1\mu_0,7,2\mu_0,10,1} F_{\mu_1,5,1\mu_1,8,1\mu_1,9,1\mu_1,9,2\mu_1,11,1} \\
& F_{\mu_2,6,1\mu_2,6,2\mu_2,9,1\mu_2,10,1\mu_2,10,2} F_{\mu_3,7,1\mu_3,8,1\mu_3,8,2\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_4,7,1\mu_4,8,1\mu_4,10,1\mu_4,11,1\mu_4,11,2} F_{\mu_5,6,1\mu_5,6,2\mu_5,10,1}^{\mu_0,5,1\mu_1,5,1} \\
& F_{\mu_0,6,1\mu_2,6,1\mu_2,6,2\mu_5,6,1\mu_5,6,2} F_{\mu_7,11,1}^{\mu_0,7,1\mu_0,7,2\mu_3,7,1\mu_4,7,1} \\
& F_{\mu_8,11,1}^{\mu_1,8,1\mu_3,8,1\mu_3,8,2\mu_4,8,1} F_{\mu_1,9,1\mu_1,9,2\mu_2,9,1\mu_3,9,1\mu_3,9,2} \\
& F_{\mu_0,10,1\mu_2,10,1\mu_2,10,2\mu_4,10,1\mu_5,10,1} F_{\mu_1,11,1\mu_4,11,1\mu_4,11,2\mu_7,11,1\mu_8,11,1}
\end{aligned}$$

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$$\begin{aligned}
& F_{\mu_0,4,1\mu_0,4,2\mu_0,6,1\mu_0,6,2\mu_0,8,1} F_{\mu_1,5,1\mu_1,5,2\mu_1,7,1\mu_1,8,1\mu_1,9,1} \\
& F_{\mu_2,7,1\mu_2,7,2\mu_2,9,1\mu_2,10,1\mu_2,11,1} F_{\mu_3,9,1\mu_3,9,2\mu_3,10,1\mu_3,10,2\mu_3,11,1} \\
& F_{\mu_4,6,1\mu_4,9,1\mu_4,10,1} F_{\mu_5,7,1\mu_5,8,1\mu_5,11,1}^{\mu_0,4,1\mu_0,4,2} \\
& F_{\mu_6,10,1\mu_6,11,1}^{\mu_0,6,1\mu_0,6,2\mu_4,6,1} F_{\mu_7,8,1}^{\mu_1,7,1\mu_2,7,1\mu_2,7,2\mu_5,7,1} \\
& F_{\mu_8,11,1}^{\mu_0,8,1\mu_1,8,1\mu_5,8,1\mu_7,8,1} F_{\mu_1,9,1\mu_2,9,1\mu_3,9,1\mu_3,9,2\mu_4,9,1} \\
& F_{\mu_2,10,1\mu_3,10,1\mu_3,10,2\mu_4,10,1\mu_6,10,1} F_{\mu_2,11,1\mu_3,11,1\mu_5,11,1\mu_6,11,1\mu_8,11,1}
\end{aligned}$$

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$$\begin{aligned}
& F_{\mu_{0,4,1}\mu_{0,6,1}\mu_{0,6,2}\mu_{0,10,1}\mu_{0,11,1}} F_{\mu_{1,5,1}\mu_{1,7,1}\mu_{1,7,2}\mu_{1,8,1}\mu_{1,9,1}} \\
& F_{\mu_{2,7,1}\mu_{2,9,1}\mu_{2,9,2}\mu_{2,10,1}\mu_{2,11,1}} F_{\mu_{3,8,1}\mu_{3,9,1}\mu_{3,10,1}\mu_{3,10,2}\mu_{3,11,1}} \\
& F_{\mu_{0,4,1}}^{\mu_{4,6,1}\mu_{4,6,2}\mu_{4,10,1}\mu_{4,11,1}} F_{\mu_{5,7,1}\mu_{5,8,1}\mu_{5,8,2}\mu_{5,9,1}}^{\mu_{1,5,1}} \\
& F_{\mu_{0,6,1}\mu_{0,6,2}\mu_{4,6,1}\mu_{4,6,2}}^{\mu_{6,11,1}} F_{\mu_{1,7,1}\mu_{1,7,2}\mu_{2,7,1}\mu_{5,7,1}}^{\mu_{7,8,1}} \\
& F_{\mu_{1,8,1}\mu_{3,8,1}\mu_{5,8,1}\mu_{5,8,2}\mu_{7,8,1}}^{\mu_{1,8,1}} F_{\mu_{1,9,1}\mu_{2,9,1}\mu_{2,9,2}\mu_{3,9,1}\mu_{5,9,1}}^{\mu_{1,9,1}} \\
& F_{\mu_{0,10,1}\mu_{2,10,1}\mu_{3,10,1}\mu_{3,10,2}\mu_{4,10,1}}^{\mu_{0,11,1}} F_{\mu_{0,11,1}\mu_{2,11,1}\mu_{3,11,1}\mu_{4,11,1}\mu_{6,11,1}}^{\mu_{0,11,1}}
\end{aligned}$$